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CLASS TEST

① Definition: Image enhancement is the process of improving the visual quality of an image by modifying its pixel values directly.

Techniques:

1. Gray-Level Transformation:

- * Changes pixel intensity values.

- * Examples: Negative, log transformation, Gamma Correction

2. Histogram processing:

- * Improves image contrast by redistributing pixel intensities.

- * Histogram Equalization is the most common method.

3. Spatial Filtering:

- * Uses a mask to process neighbouring pixels.

- * Smoothing filters remove noise.

Applications:

- * Medical Imaging

- * Face recognition

- * Industrial inspection

② Image Smoothing:

- * Removes noise from an image
- * Uses Low-pass filters
- * Produces a blurred image
- * Examples: Mean, Median and Gaussian filters.

Image Sharpening:

- * Enhances edges and fine details.
- * Uses High-pass filters.
- * Produces a sharper image.
- * Examples: Laplacian, Sobel and unsharp masking.

Difference

Smoothing

~~Removes~~ noise
~~Blurs~~ image
Low-pass filter

Sharpening

Enhances edges
Sharpening image
High-pass filter.

Applications:

- * Smoothing: Medical images, noise reduction.
- * Sharpening: Object detection, OCR, face recognition.

Definition:

Frequency domain enhancement improves image quality by modifying the frequency components of an image using the Fourier Transform.

Process:

- Convert image into frequency domain.
- Apply frequency filter.
- Convert back using Inverse Fourier Transform.

Types of Filters

Low-Pass Filter

- * Removes high-frequency noise.
- * Produces smooth images.

High-Pass Filter

- * Enhances high-frequency components.
- * Improves edges and image sharpening.

Applications:

- Noise removal
- Medical imaging
- Satellite image processing
- Edge Enhancement.

4) Defination:

Image segmentation divides image into meaningful regions for easier analysis.

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Techniques:

- * Point Detection: Detects isolated pixels
- * Line Detection: Detects horizontal, vertical and diagonal lines.
- * Edge Detection: Detect object boundaries.
- * Thresholding: separates foreground and background using intensity values.
- * Region-Based Segmentation: Groups neighboring pixels with similar properties.

Applications:

- * Medical imaging
- * Face recognition
- * OCR
- * Object detection.
- * Satellite image analysis

Q. Edge Detection:

Detects sudden changes in pixel intensity to identify object boundaries.

Examples: Sobel, Prewitt, Roberts and Canny operators

Edge Linking:

Connects broken or disconnected edges to form continuous boundaries.

Boundary detection:

Extracts the complete outline of an object after edge linking.

Applications:

- Face recognition
- Medical image analysis
- Industrial quality inspection
- Autonomous vehicles
- Object tracking

Conclusion: These three techniques work together to identify, connect and extract object boundaries for image analysis.